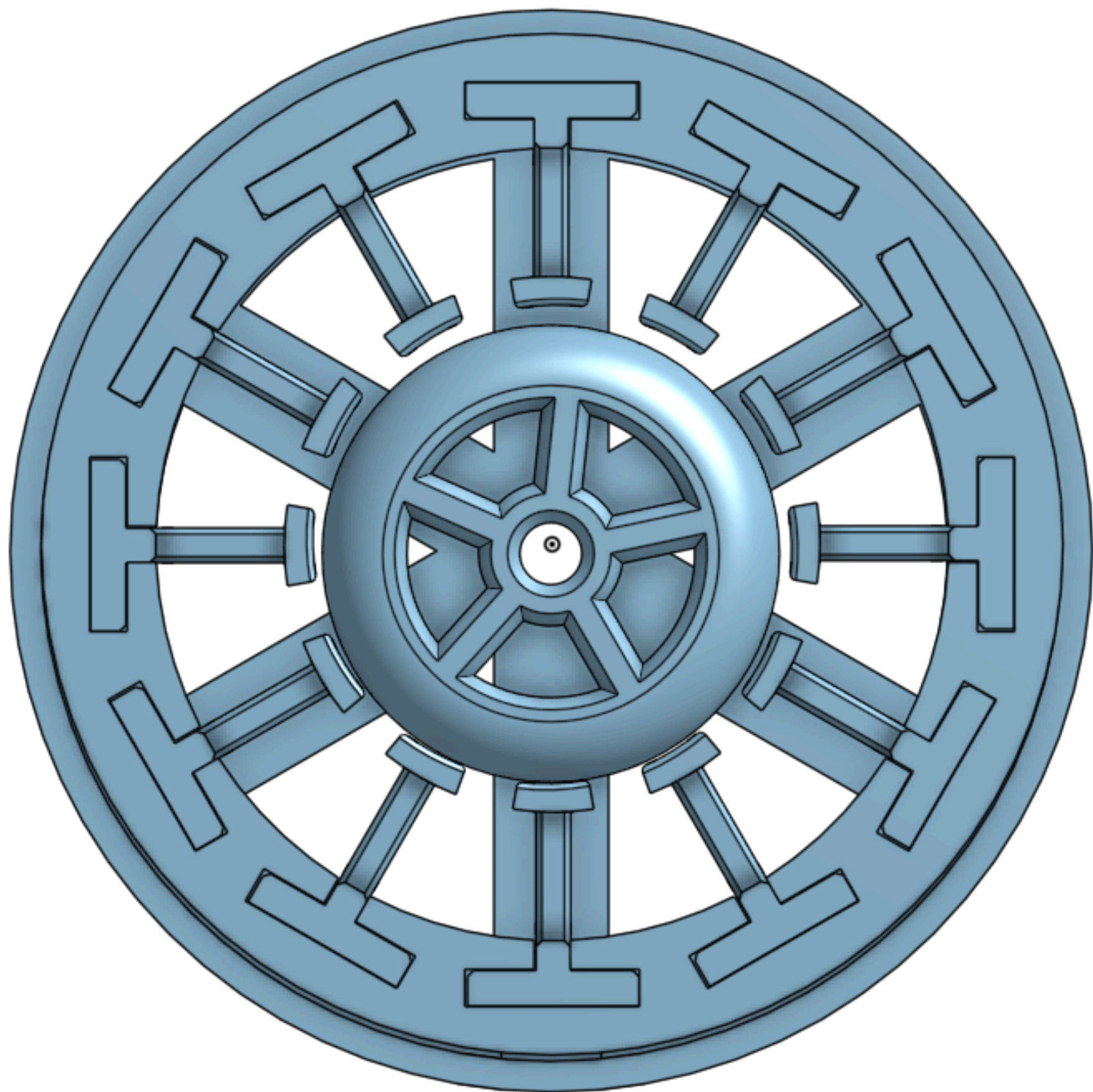




BRUSHLESS DC MOTOR

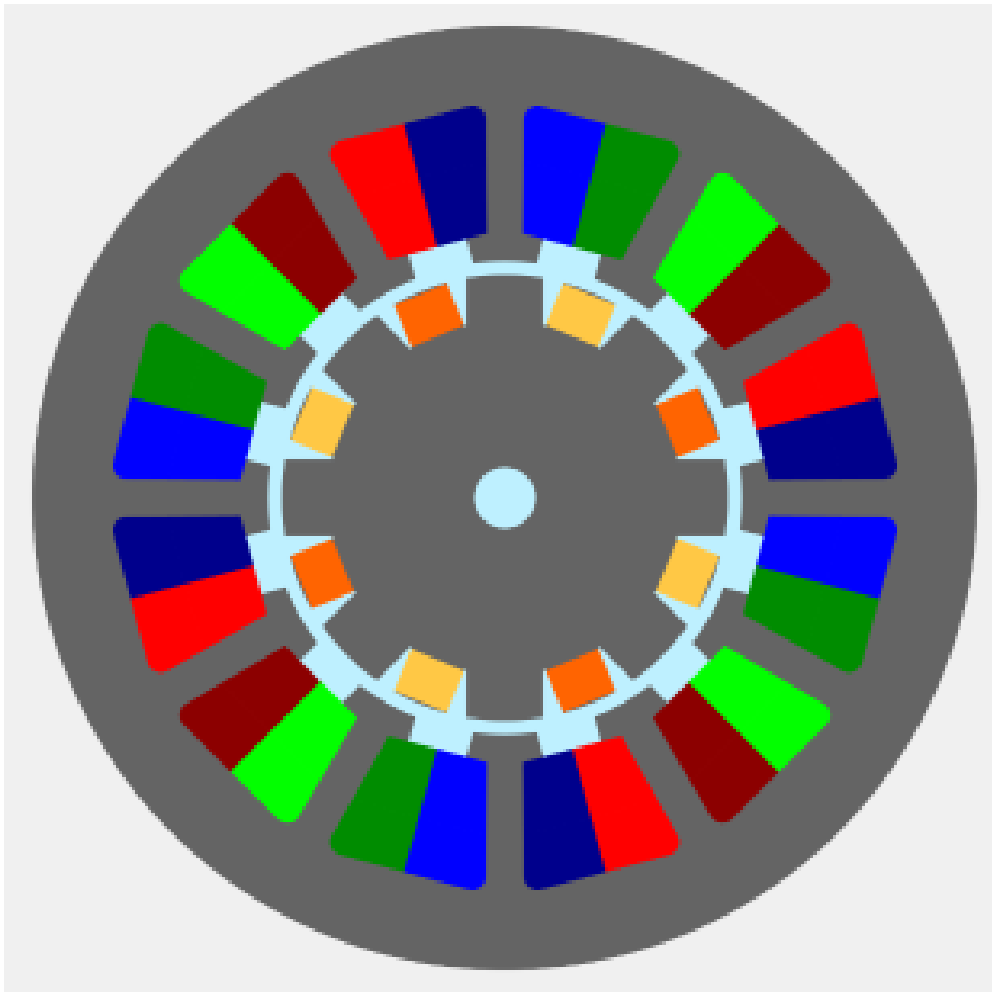
Understanding How a BLDC Motor Works

A Brushless DC (BLDC) motor operates through the interaction of a rotating magnetic field in the stator and permanent magnets in the rotor, generating torque based on the Lorentz Force. Unlike brushed motors, it uses electronic commutation, making it more efficient, reliable, and low maintenance.

- DC supply is converted into a 3-phase supply using an ESC (Electronic Speed Controller)
- The 3-phase supply is fed to the stator (Y-connected 3-phase windings), creating a rotating magnetic field (RMF)
- The rotor contains strong permanent neodymium magnets, whose magnetic field aligns with the RMF, resulting in torque generation



Topology of motor



16 Slot 8pole Inner Rotor

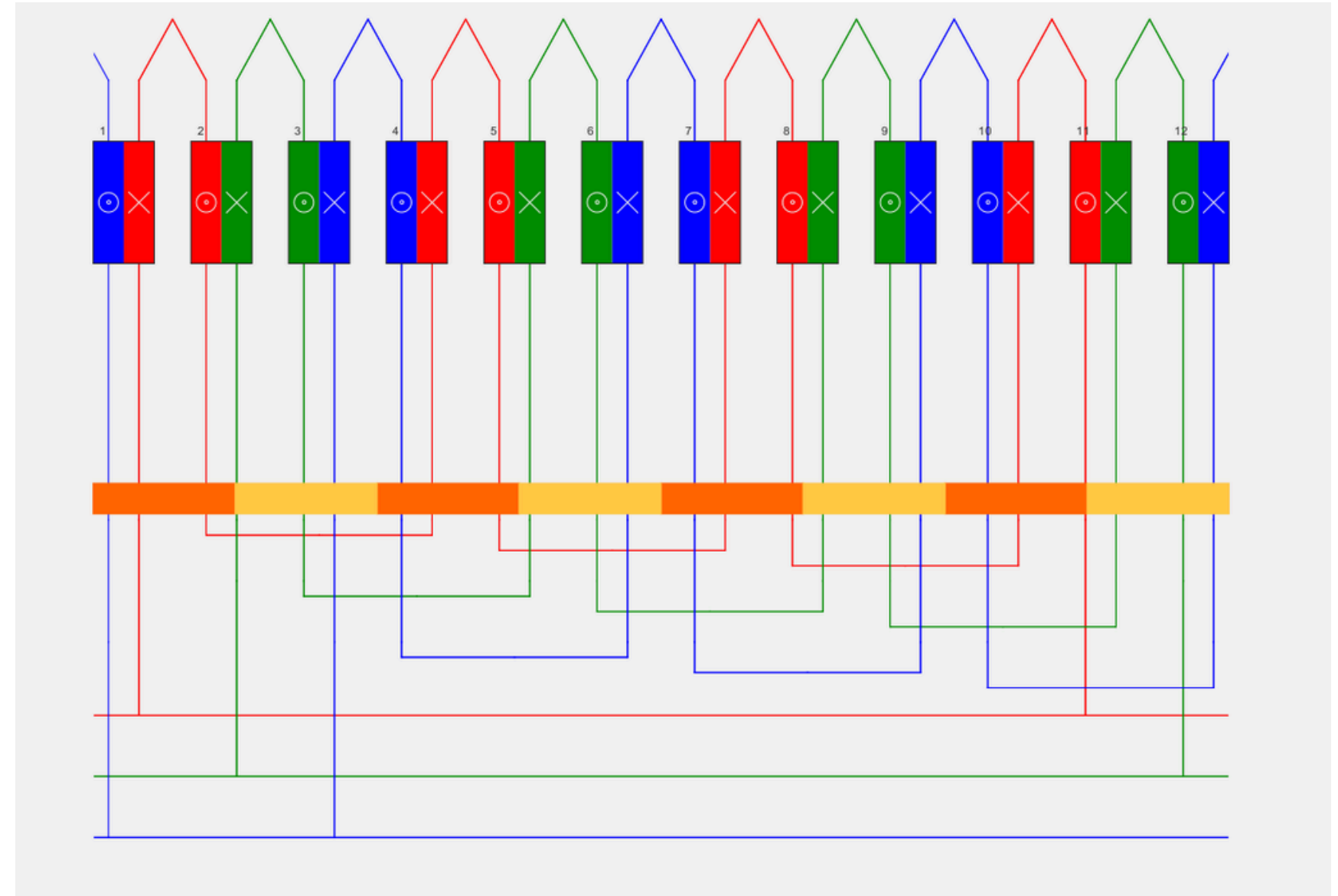
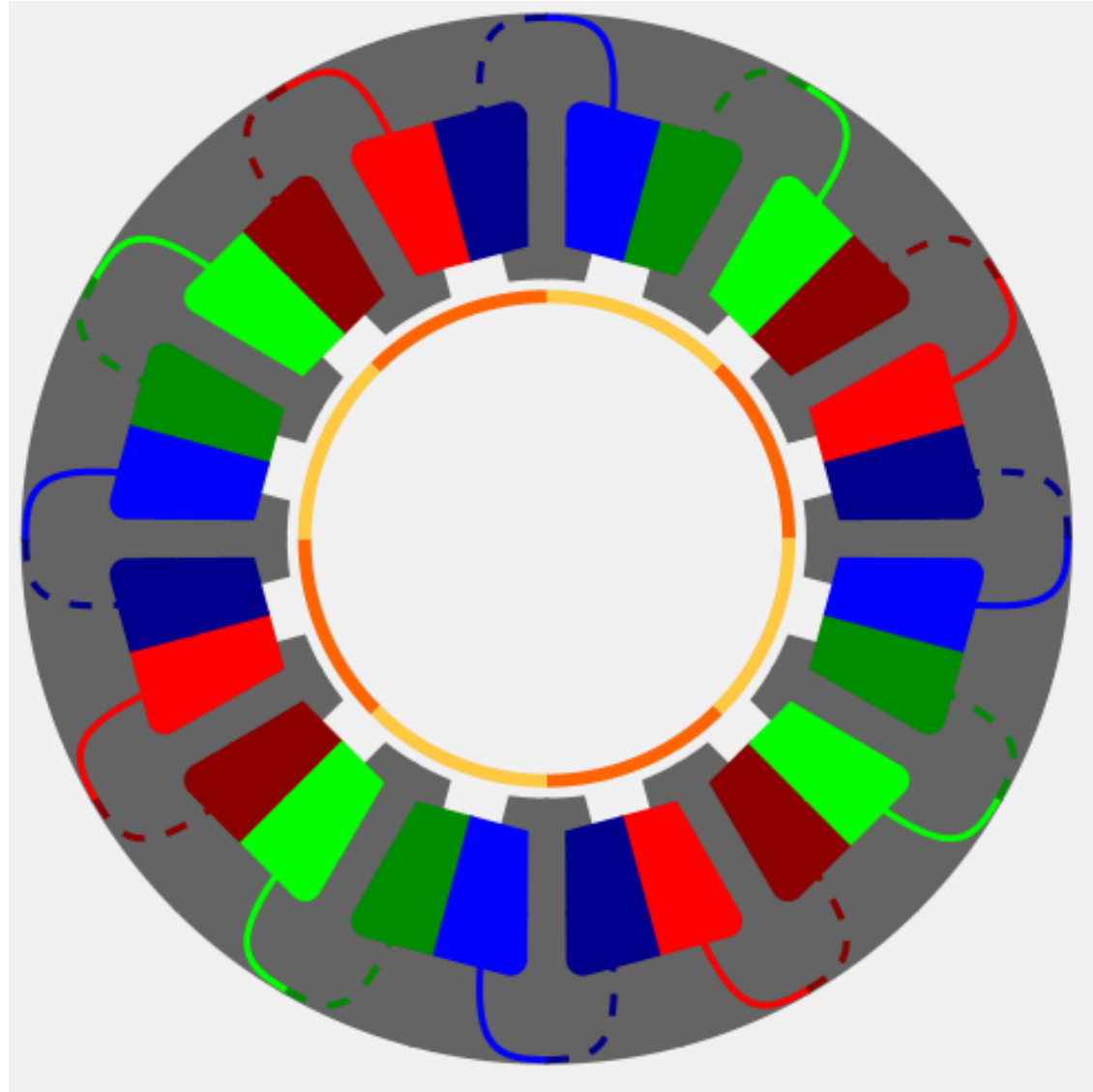
Stator	
Outer diameter (mm)	80.0
Inner diameter (mm)	40.0
Length (mm)	22.0
No. slots	12

Airgap	
Length (mm)	1.0

Rotor	
Outer diameter (mm)	38.0
Inner diameter (mm)	5.0
Length (mm)	22.0
No. poles	8

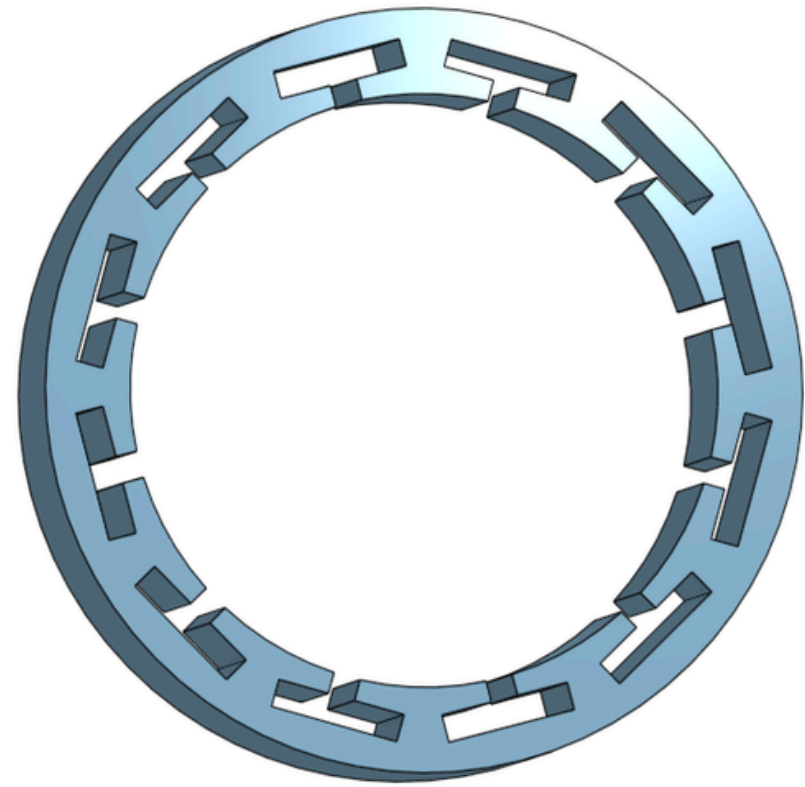
Stator and Rotor Parameters

Stator Winding

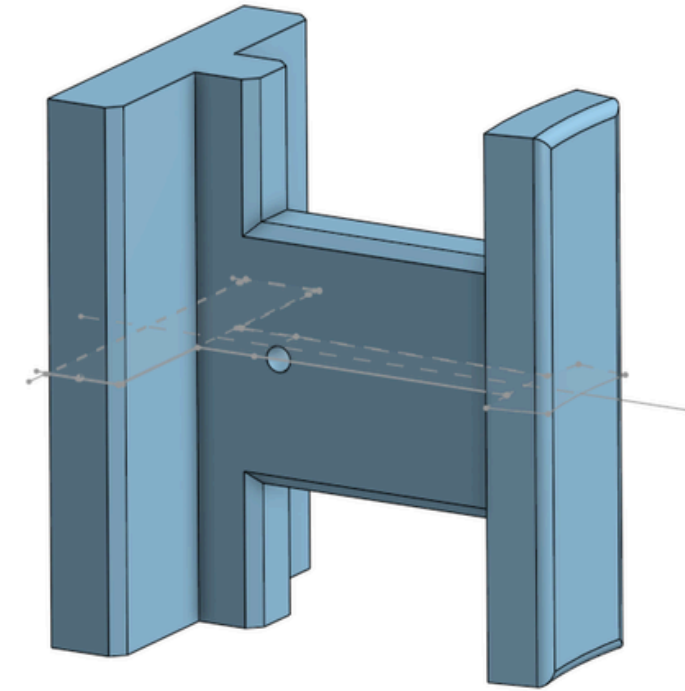
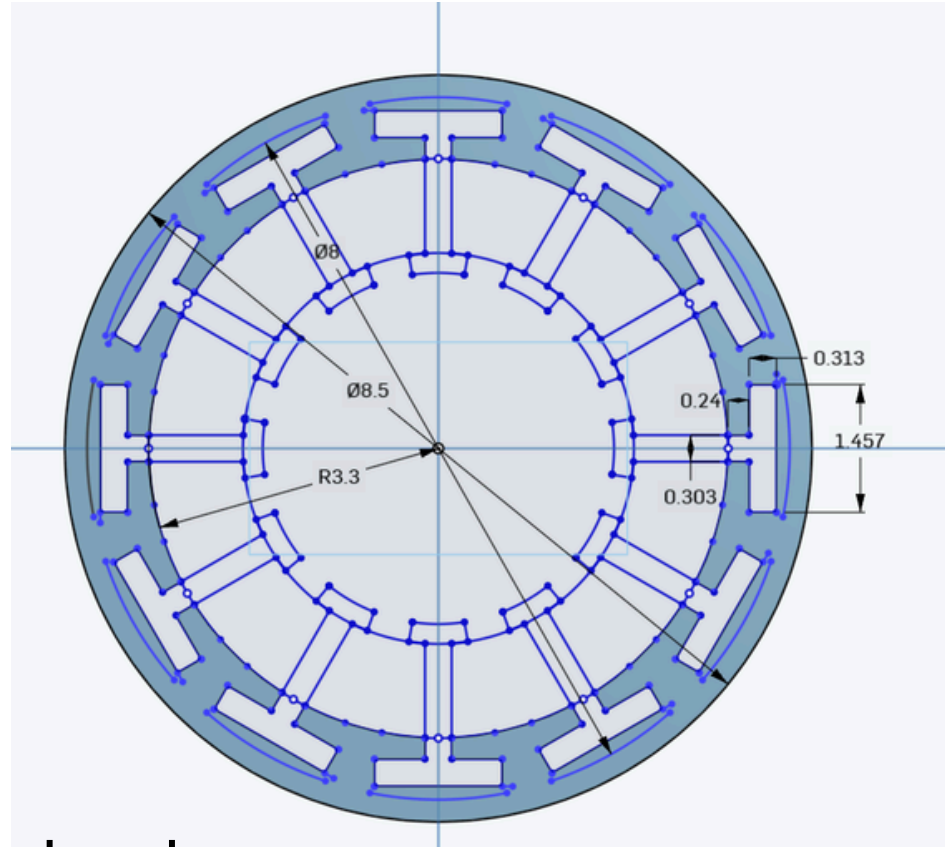


- Enamelled copper wire with a diameter of 0.5 mm was utilized.
- Each slot contains 50 coils.
- The coils are arranged in a **STAR** connection.
- Winding Factor = 0.866.

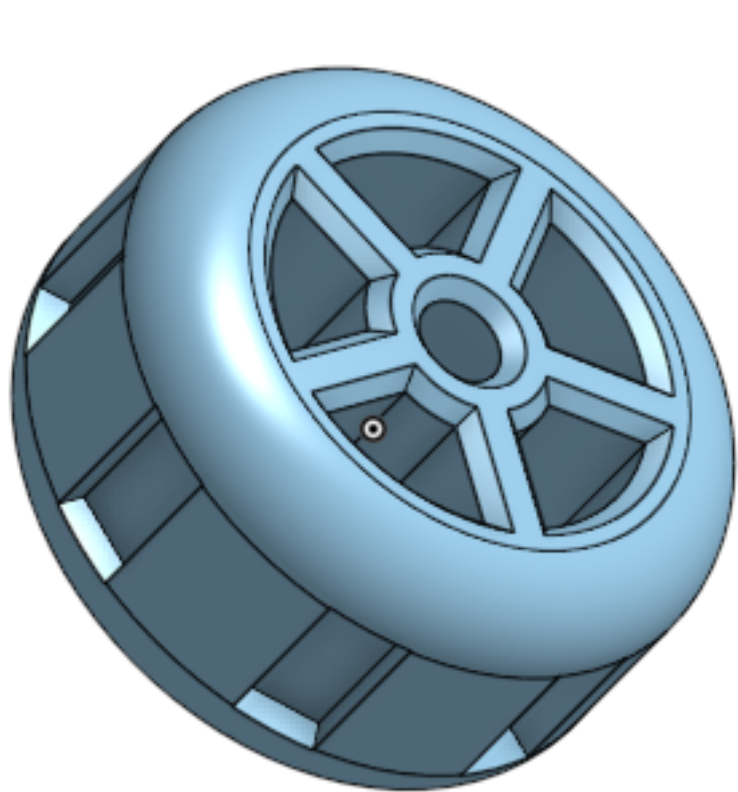
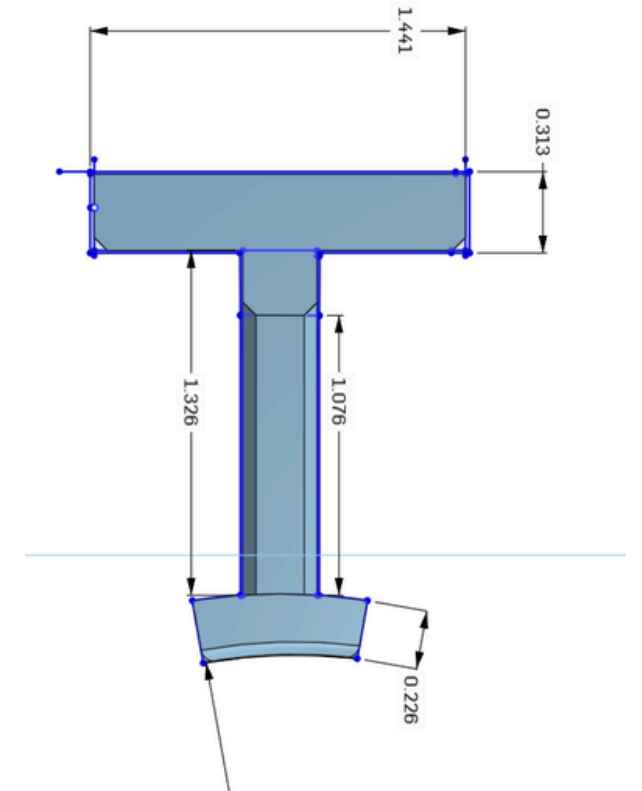
CAD Design



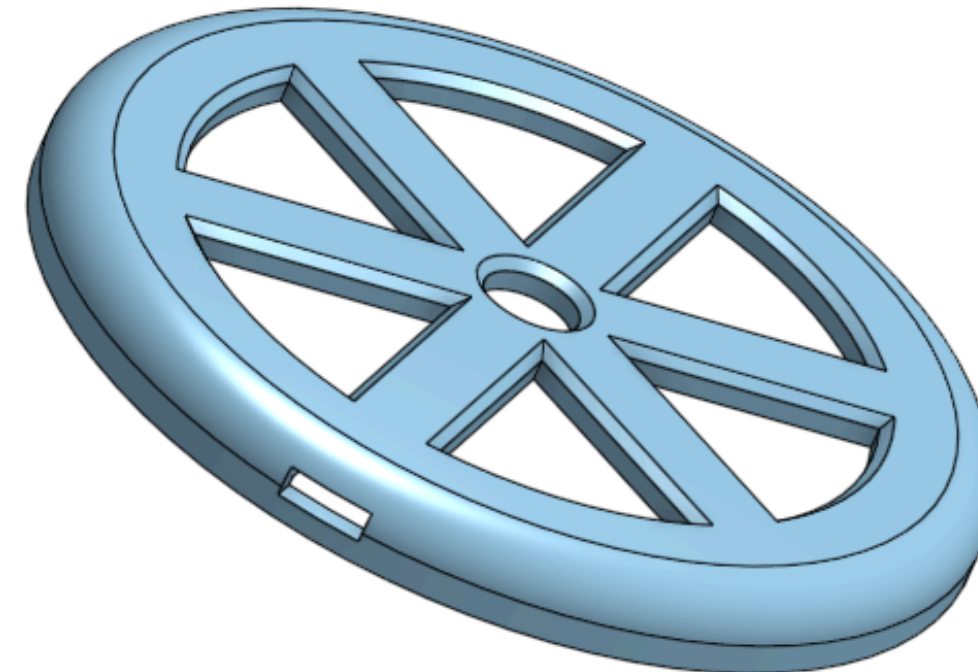
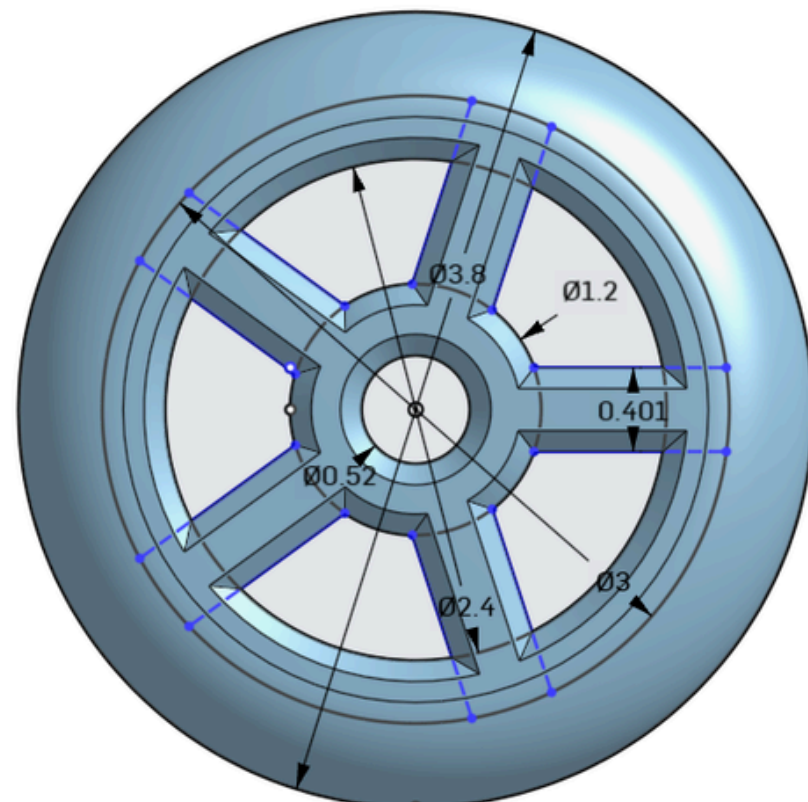
Stator body



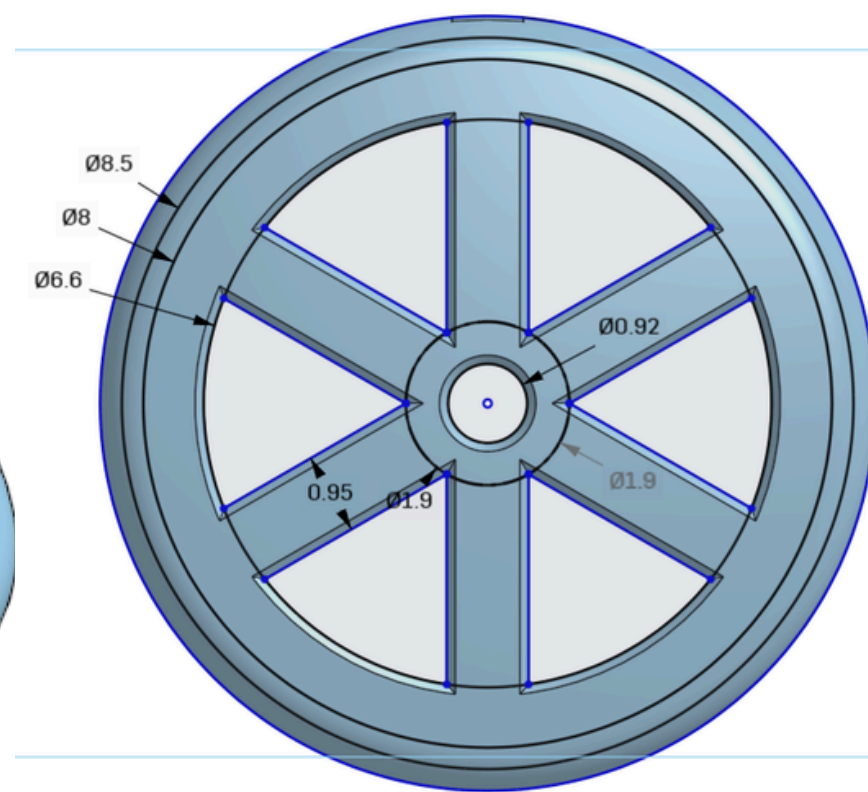
Stator teeth



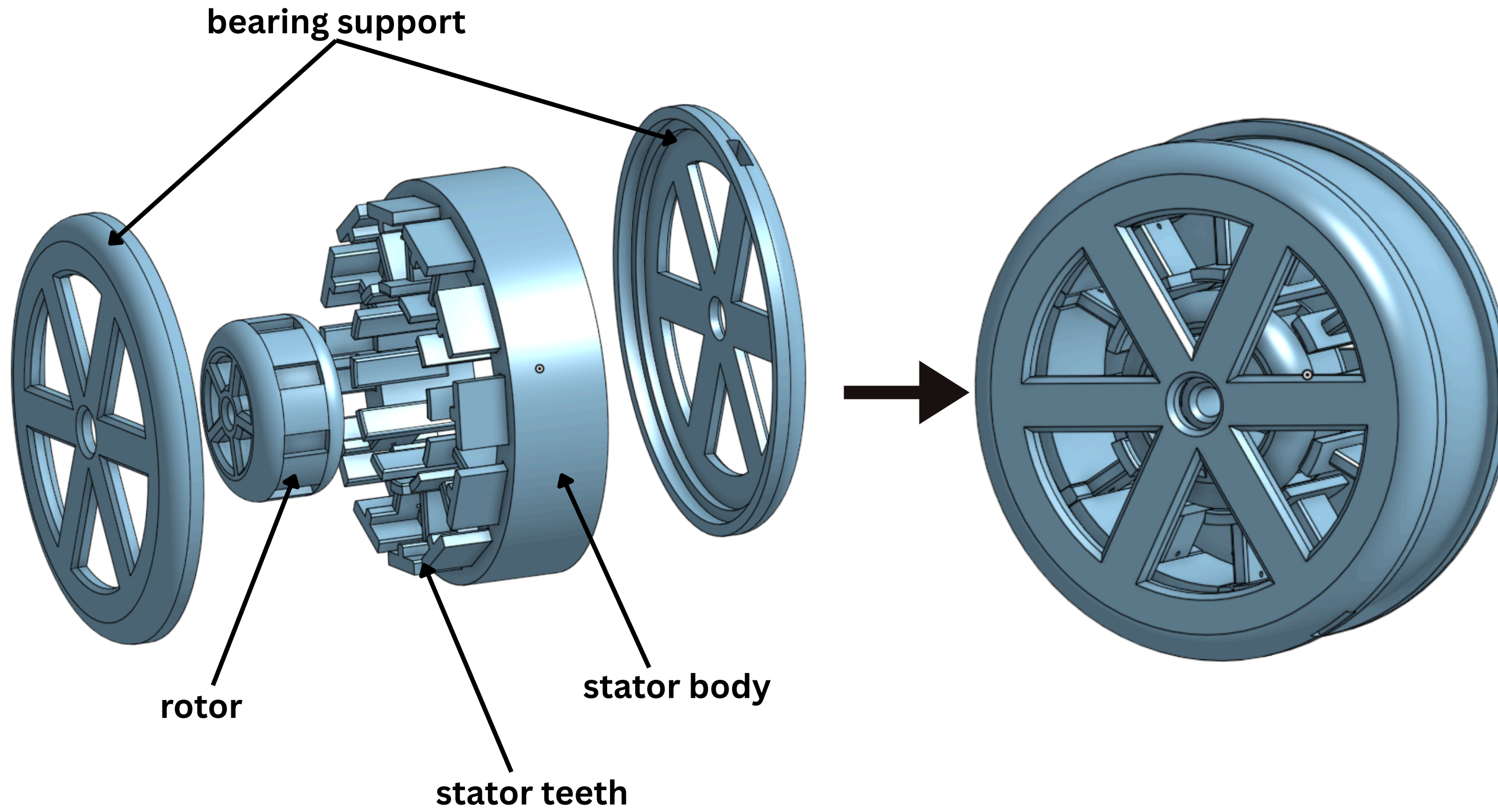
Rotor body



Bearing support



Assembly



Simulation

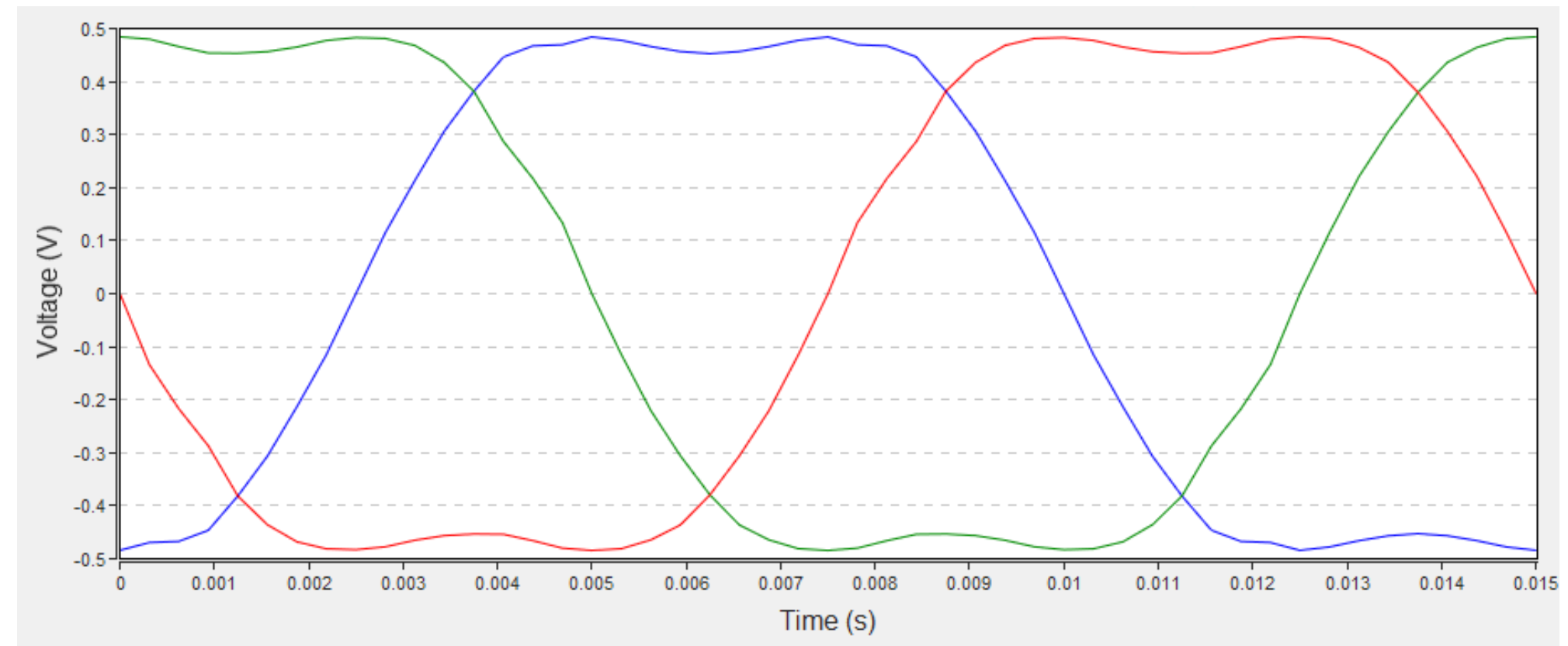
Back emf test

BLDC motors have a characteristic trapezoidal back EMF waveform, which occurs because the stator winding interacts with rectangular permanent magnets on the rotor

- Flat region (under North pole)
- Sudden drop (at the edge of the magnet)
- Flat region (under South pole)
- Sudden rise (at the edge of the magnet)

This results in a quasi-trapezoidal back EMF waveform, which is typical for BLDC motors.

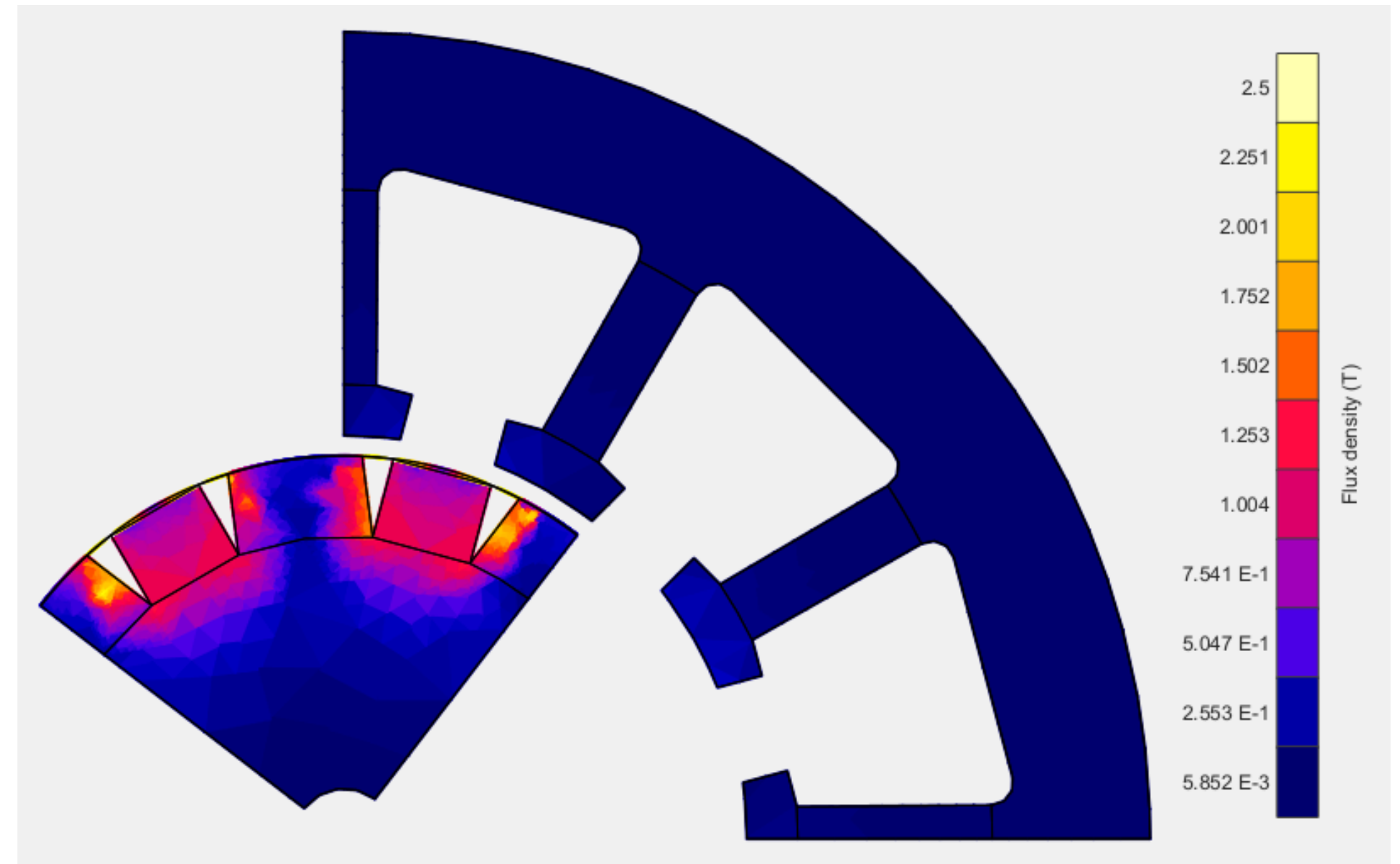
The small ripples observed on the waveform are due to harmonics that combine (via Fourier series) to form the final trapezoidal shape.



Phase voltage versus time - Open circuit
Test speed = 1000RPM

Flux Density Due to Rotor Magnets

- Displays air-gap flux density distribution in open-circuit condition
- Represents magnetic field from rotor permanent magnets
- Higher flux areas indicate magnet pole locations
- Lower flux areas are found in air/slot regions between poles
- Distribution non-uniformity results from stator slotting and geometry



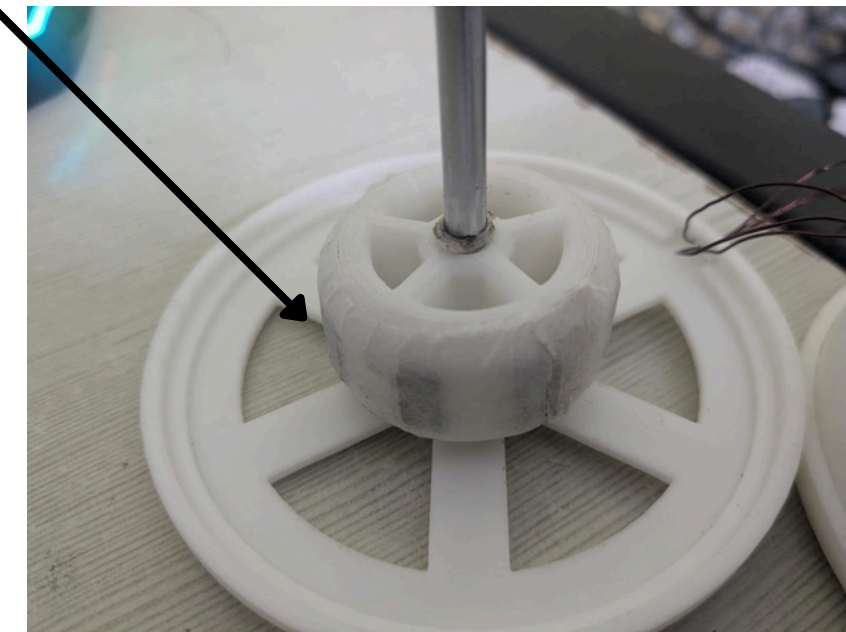
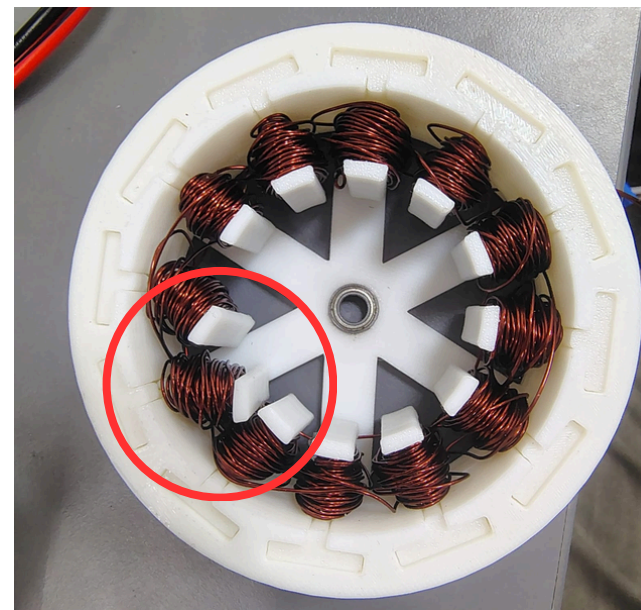
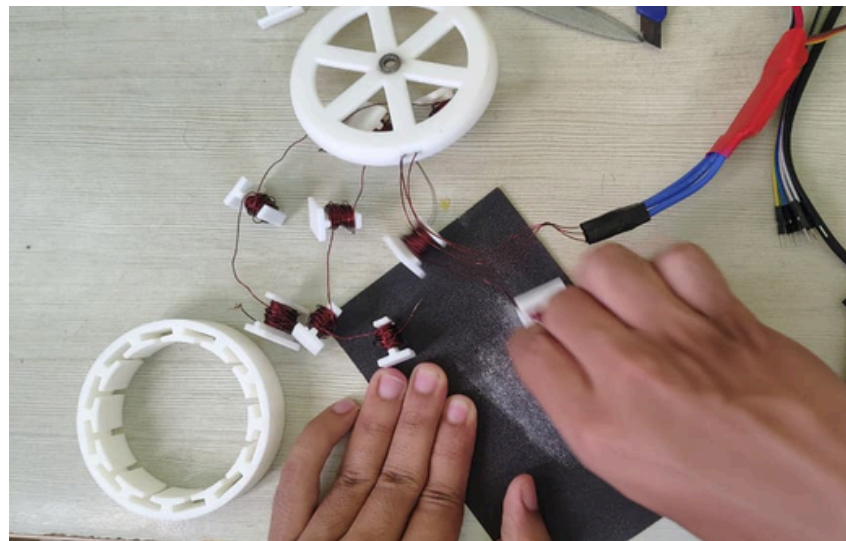
Isovalues

Challenges faced and solutions implemented

- Differences in CAD model and particle 3D model tolerances led to repeated breakage of stator slots
- Low 3D fill (15%) resulted in a lighter rotor, which caused increased instability and wobbling
- At high speeds, rotor magnets dislodged, resulting in damage to the stator
- Selection of appropriate vendor for components

SOLUTION

- Sanded the stator's teeth to create additional space for the rotor and mitigate vibrations
- Secured the magnets using superglue and added tape to minimize the risk of them coming loose



Softwares used

- Onshape
- Altair Flux Motor 2026
- Bambu Studio